

Sean Luka Girgis

Senior Data Engineer | Python, SQL, ETL/ELT, Data Lakes & Warehouses

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Professional Summary

Senior Data Engineer with 20+ years of enterprise technology experience across Python, SQL, PySpark/Spark, ETL/ELT pipelines, data lake and warehouse patterns, data marts, APIs, XML/JSON, data modeling, query performance tuning, production support, documentation, and technical mentorship. Strong background designing and supporting scalable data workflows, validating reporting outputs, optimizing SQL/database layers, building analytics-ready datasets, and translating business requirements into reliable data solutions. Best aligned to senior data engineering roles requiring Python/SQL, pipeline design, data architecture, Airflow/orchestration concepts, API-based ingestion, data quality, performance tuning, CI/CD/testing discipline, rapid POCs, and cross-functional technical leadership; Azure ADF/ADLS, Microsoft SQL Server, dbt, Snowflake, Databricks, BigQuery, SSAS/SSRS, Power BI, Tableau, and proptech domain data are positioned carefully as adjacent or transferable areas rather than overstated production ownership.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Designed and supported Python/Pandas/PySpark and SQL ETL/ELT workflows ingesting telemetry from 6,000+ infrastructure endpoints into curated datasets for reporting, analytics, forecasting, validation, and decision support.
- Built AWS and warehouse-backed data platform components using S3 landing patterns, AWS Glue, Athena, Amazon Redshift, Oracle, SQL, and PySpark to support scalable ingestion, transformation, querying, and analytics.
- Created data lake, warehouse, and data mart-style reporting structures that transformed raw operational feeds into reusable, validated datasets for technical and business stakeholders.
- Used complex SQL across Oracle, Amazon Redshift, and reporting structures to validate transformations, troubleshoot discrepancies, tune queries, and improve downstream reporting performance.
- Implemented data quality and reconciliation checks for row counts, nulls, duplicates, unmatched joins, freshness, abnormal deltas, and output comparisons against trusted prior runs.
- Worked with infrastructure, application, analytics, engineering, and business teams to clarify requirements, document metric definitions, explain data behavior, prioritize fixes, and deliver trusted outputs.
- Created runbooks, source mappings, conceptual data flows, metric definitions, validation checkpoints, support procedures, and technical documentation to improve maintainability and handoff.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Built analytics and reporting outputs from Dynatrace AppMon and Gomez Synthetic Monitoring data for Brand.com and critical hospitality systems.
- Analyzed real-user and synthetic monitoring datasets to identify data quality issues, performance trends, and optimization opportunities.

- Created dashboards and reporting views that converted raw monitoring data into actionable performance, reliability, and stakeholder insights.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Supported a large financial-services monitoring and reporting environment with 50+ Enterprise Managers and 4,000–6,000 instrumented agents.
- Built Perl and KornShell automation for extracting, validating, transforming, and distributing APM telemetry, replacing manual reporting workflows with repeatable data-processing pipelines.
- Designed dashboards, alert rules, SLA thresholds, reporting standards, and documentation used by application, middleware, infrastructure, and operations teams.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Developed and executed performance test plans for enterprise telecom applications, including baseline comparisons, regression review, and release-readiness analysis.
- Analyzed application behavior under load to identify throughput limits, SQL/JDBC bottlenecks, CPU, memory, thread, and garbage-collection issues.
- Documented test results, expected behavior, defects, risks, and remediation recommendations for engineering and release teams.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster while preserving transaction performance and data integrity.
- Built C++/OCCI/OCI validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.

Education

Post-Graduate Diploma in Computer Engineering Technology / Computer Science

Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering

Zagazig University, Egypt

Skills

Languages: Python, SQL, PySpark, PL/SQL, XML, JSON, Perl, KornShell/ksh, C++, C, Java/Scala concepts

Flagship Projects

Serverless Lakehouse Platform (AWS)

Designed an AWS data platform pattern using S3, Glue, Athena, Redshift, PySpark, and Parquet for raw ingestion, curated transformations, analytics datasets, query optimization, and validation checkpoints.

Technologies: AWS S3, AWS Glue, Athena, Redshift, PySpark, Parquet, SQL

Production Data Quality and RCA Workflow

Built validation and reconciliation workflows for production reporting datasets, including row-count checks,

null checks, duplicate/unmatched record checks, missing feed diagnosis, and prior-run comparisons.

Technologies: SQL, Python, Pandas, Data Validation, RCA, Runbooks

HorizonScale — Capacity Forecasting and Data Platform

Built Python/Pandas/PySpark workflows to process telemetry, validate outputs, prepare forecasting datasets, and produce trusted reporting for capacity and risk discussions.

Technologies: Python, Pandas, PySpark, SQL, Prophet, Data Validation

ServiceCall AI RAG Demo

Built staged Python proof-of-concepts with document loading, chunking, TF-IDF retrieval, hybrid retrieval, Pydantic schemas, FastAPI service layers, Docker-first smoke tests, and deterministic API endpoints.

Technologies: Python, FastAPI, Docker, Pydantic, pytest, RAG, TF-IDF